

- Windows as the host OS
 - VMware workstation (Any guest OS)
 - VirtualBox (Any guest OS)
 - Hyper-V (Any guest OS)
- Linux as the host OS
 - VMware workstation
 - Microsoft virtual PC
 - VMLite workstation
 - VirtualBox
 - Xen

A hypervisor or virtual machine monitor (VMM) is a piece of computer software, firmware or hardware that creates and runs virtual machines. A computer on which a hypervisor is running one or more VMs is defined as a host machine. Each VM is called a guest machine. The hypervisor presents the guest OSs with a virtual operating platform, and manages the execution of the guest operating systems. Multiple instances of a variety of operating systems may share the virtualised hardware resources.

Hypervisors of Type 1 (bare metal installation) and Type 2 (hosted installation)

When implementing and deploying a cloud service, Type 1 hypervisors are used. These are associated with the concept of bare metal installation. It means there is no need of any host operating system to install the hypervisor. When using this technology, there is no risk of corrupting the host OS. These hypervisors are directly installed on the hardware without the need for any other OS. Multiple VMs are created on this hypervisor.

A Type 1 hypervisor is a type of client hypervisor that interacts directly with hardware that is being virtualised. It is completely independent of the operating system, unlike a Type 2 hypervisor, and boots before the OS. Currently, Type 1 hypervisors are being used by all the major players in the desktop virtualisation space, including but not limited to VMware, Microsoft and Citrix.

The classical virtualisation software or Type 2 hypervisor is always installed on any host OS. If the host OS gets corrupt or crashes for any reason, the virtualisation software or Type 2 hypervisor will also crash and, obviously, all VMs and other resources will be lost. That's why the hypervisor technology or bare metal installation is very popular in the cloud computing world.

Type 2 (hosted) hypervisors execute within a conventional OS environment. With the hypervisor layer as a distinct second software level, guest OSs run at the third level above the hardware. A Type 2 hypervisor is a type of client hypervisor that sits on top of an OS. Unlike a Type 1 hypervisor, a Type 2 hypervisor relies heavily on the operating system. It cannot boot until the OS is already up and running, and if for any reason the OS crashes, all end users are affected. This is a big drawback of Type 2 hypervisors, as they are only as secure as the OS on which they rely. Also,

since Type 2 hypervisors depend on an OS, they are not in full control of the end user's machine.

Hypervisor Type 1 products

- VMware ESXi
- Citrix Xen
- KVM (Kernel Virtual Machine)
- Hyper-V

Hypervisor Type 2 products

- VMware Workstation
- VirtualBox

Table 1
Hypervisors and their cloud service providers

Hypervisor	Cloud service provider
Xen	Amazon EC2 IBM SoftLayer Fujitsu Global Cloud Platform Linode OrionVM
ESXi	VMware Cloud
KVM	Red Hat HP Dell Rackspace
Hyper-V	Microsoft Azure

Data centres and uptime tier levels

Just as a virtual machine is mandatory for cloud computing, the data centre is also an essential part of the technology. All the cloud computing infrastructure is located in remote data centres where resources like computer systems and associated components, such as telecommunications and storage systems, reside. Data centres typically include redundant or backup power supplies, redundant data communications connections, environmental controls, air conditioning, fire suppression systems as well as security devices.

The tier level is the rating or evaluation aspect of the data centres. Large data centres are used for industrial scale operations that can use as much electricity as a small town. The standards comprise a four-tiered scale, with Tier 4 being the most robust and full-featured (Table 2).

Cloud simulations

Cloud service providers charge users depending upon the space or service provided.

In R&D, it is not always possible to have the actual cloud infrastructure for performing experiments. For any research scholar, academician or scientist, it is not feasible to hire cloud services every time and then execute their algorithms or implementations.

For the purpose of research, development and testing, open source libraries are available, which give the feel of cloud services. Nowadays, in the research market, cloud simulators are widely used by research scholars and practitioners, without the need to pay any amount to a cloud service provider.